

Session 1

Global landscape of data governance institutions and frameworks

Ernest Guevarra 

ernest@guevarra.io

University of Oxford

Definitions

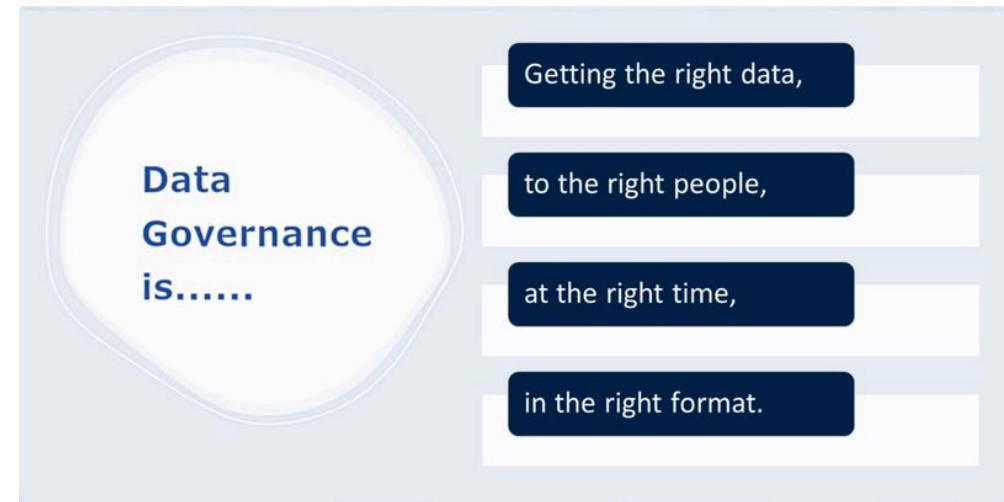
What is data?

“When we refer to data, we mean information about **people, things** and **systems**. ... Data about people can include **personal data**, such as *basic contact details*, records generated through *interaction with services* or the *web*, or information about their *physical characteristics (biometrics)* - and it can also extend to **population-level data**, such as *demographics*. Data can also be about **systems and infrastructure**, such as *administrative records* about businesses and public services. Data is increasingly used to describe **location**, such as *geospatial reference* details, and the *environment* we live in, such as data about *biodiversity* or the *weather*. It can also refer to the information generated by the *burgeoning web of sensors* that make up the *Internet of Things*.”

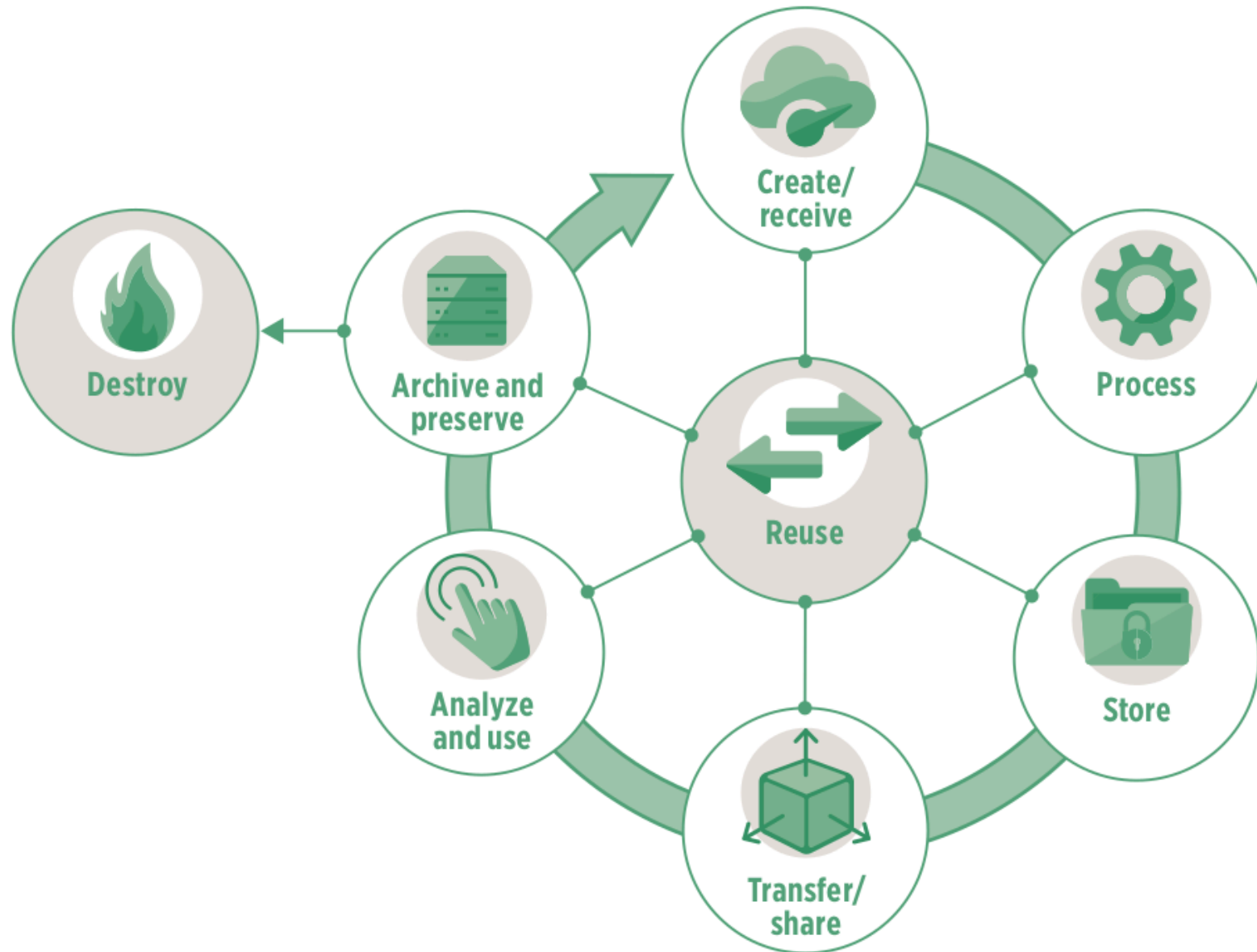
- UK National Data Strategy

What is data governance?

A system of robust legal frameworks, ethical guidelines, standards, and processes that ensures an organisation's data is accurate, consistent, secure, and compliant throughout its lifecycle.



What is the data lifecycle?



Public vs Private intent data

Public intent data

- Public intent data is data collected for public purposes.
 - Public intent data is a foundation of public policies and can play a transformative role in the public sector.
- Private intent data is data collected by the private sector as part of routine business processes.
 - Private intent data can drive innovation, lower transaction costs, and improve productivity, competitiveness, and economic growth.

Activity 1: Tell me about your data

Activity 1 - Mechanics

1. Move into the group assigned to you.

- TAMRI Hanoi + TAMRI Ho Chi Minh City
- Tam Anh Ho Chi Minh City
- Tam Anh Hanoi
- Eplus + SMART Research
- VNVC + Calif
- Department of Health (Hanoi and Ho Chi Minh City) +
Department of Science and Technology Hanoi

Activity 1 - Mechanics

2. Individually, write keywords and/or key terms/concepts that describe any one or all data that you work with and/or are in charge of

- write on the provided meta cards;
- use keywords or key phrases and not sentences;
- one keyword or key phrase in one meta card;
- you can use as many meta cards as you want/need.

Activity 1 - Mechanics

3. Post your meta cards onto the assigned wall or table for your group.
 - Make sure that your meta cards are separate from everyone else's in your group;
 - Make sure that your meta cards are posted in such a way that the text in each card is readable (don't make the cards overlap).

Activity 1 - Mechanics

4. Within your group, find a partner and discuss and describe with each other the data that you work with and/or are in charge of
- You will be given 5 minutes to share with your partner;
 - After 5 minutes, find a new partner and then discuss and describe with each other;
 - We will do three rounds of partner discussion within group.

Activity 1 - Mechanics

5. With another group, find a partner in that group and then ask them to discuss and describe to you the data they work with and/or in charge of
- Specific groups will be chosen to approach another group;
 - The members of the visiting group will find a partner in the hosting group;
 - The members of the visiting group will ask their partner from the hosting group about the data they work with and/or are in charge of. They will have about 5 minutes for this discussion;
 - After five minutes, the members of the visiting group should find a new partner among the members of the hosting group and then start asking about the data they work with and/or are in charge of;
 - After three rounds, the hosting group will then be asked to visit another group and do the same steps while the visiting group will then host a different group.

Public intent data

Types of public intent data

- Administrative data
- Census data
- Sample surveys
- Citizen-generated data
- Machine-generated data
- Geospatial data

Maximising value of public intent data

Ensuring the data have adequate coverage

- Completeness
- Timeliness
- Frequency

Ensuring the data are of high quality

- Granularity
- Accuracy
- Comparability

Ensuring the data are easy to use

- Accessibility
- Understandability
- Interoperability

Ensuring the data are safe to use

- Impartiality
- Confidentiality
- Appropriateness

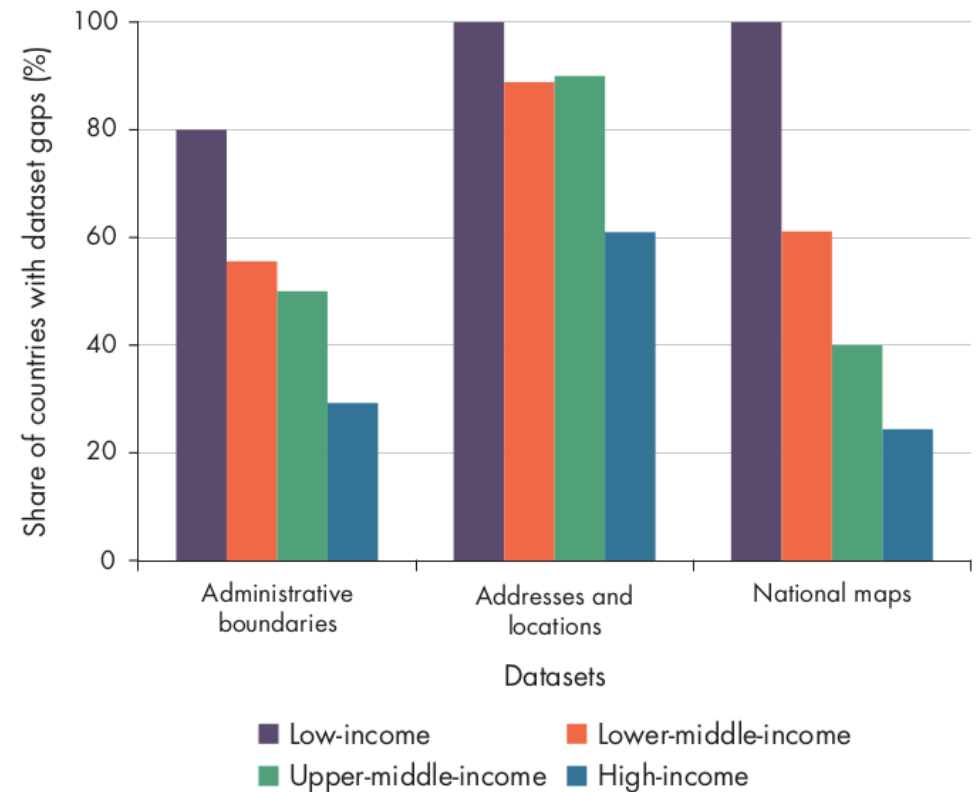
Three pathways for public intent data to add value

- Improving service delivery
- Prioritising scarce resources
- Holding government accountable and empowering individuals

Gaps in public intent data

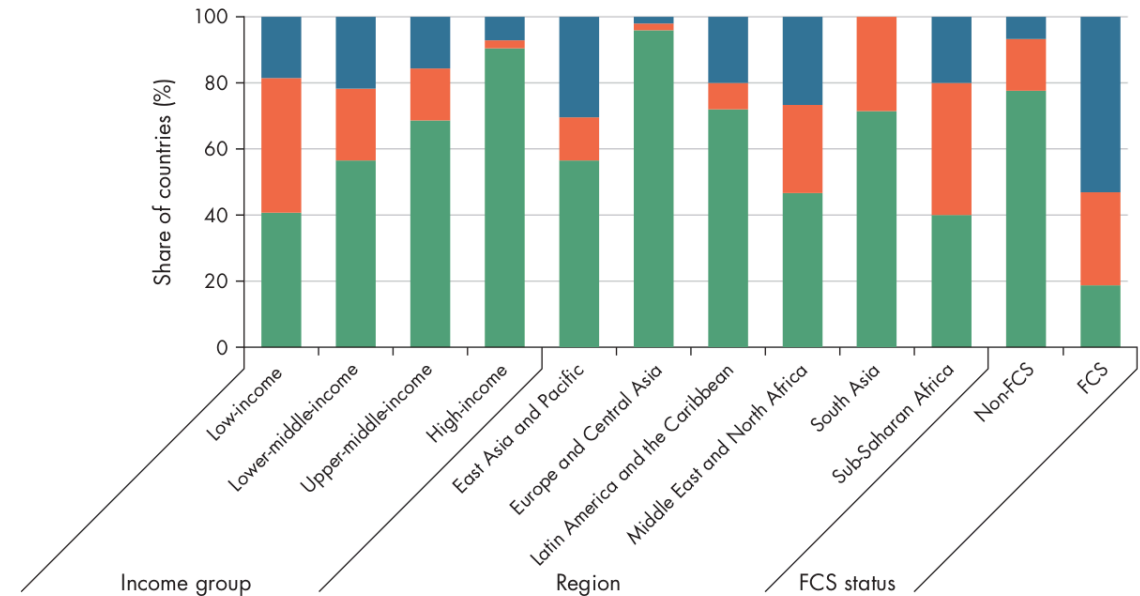
Coverage is inadequate

- Lack of *timeliness, frequency, and completeness*.
- Large gaps in geospatial data in lower-income countries



Data quality is poor

- Lack of *granularity, accuracy, and comparability*.
- Lack of data disaggregation
- Lower-income countries have less comparable poverty data



Data not easy to use

- Lack of *accessibility, understandability, and interoperability*.

Indicator	Low-income	Low-middle-income	Upper-middle-income	High-income
Openness score (0–100)	38	47	50	66
Available in machine readable format (%)	37	51	61	81
Available in non-proprietary format (%)	75	85	81	84
Download options available (%)	56	68	68	78
Open terms of use/license (%)	11	19	22	44

Data not safe to use

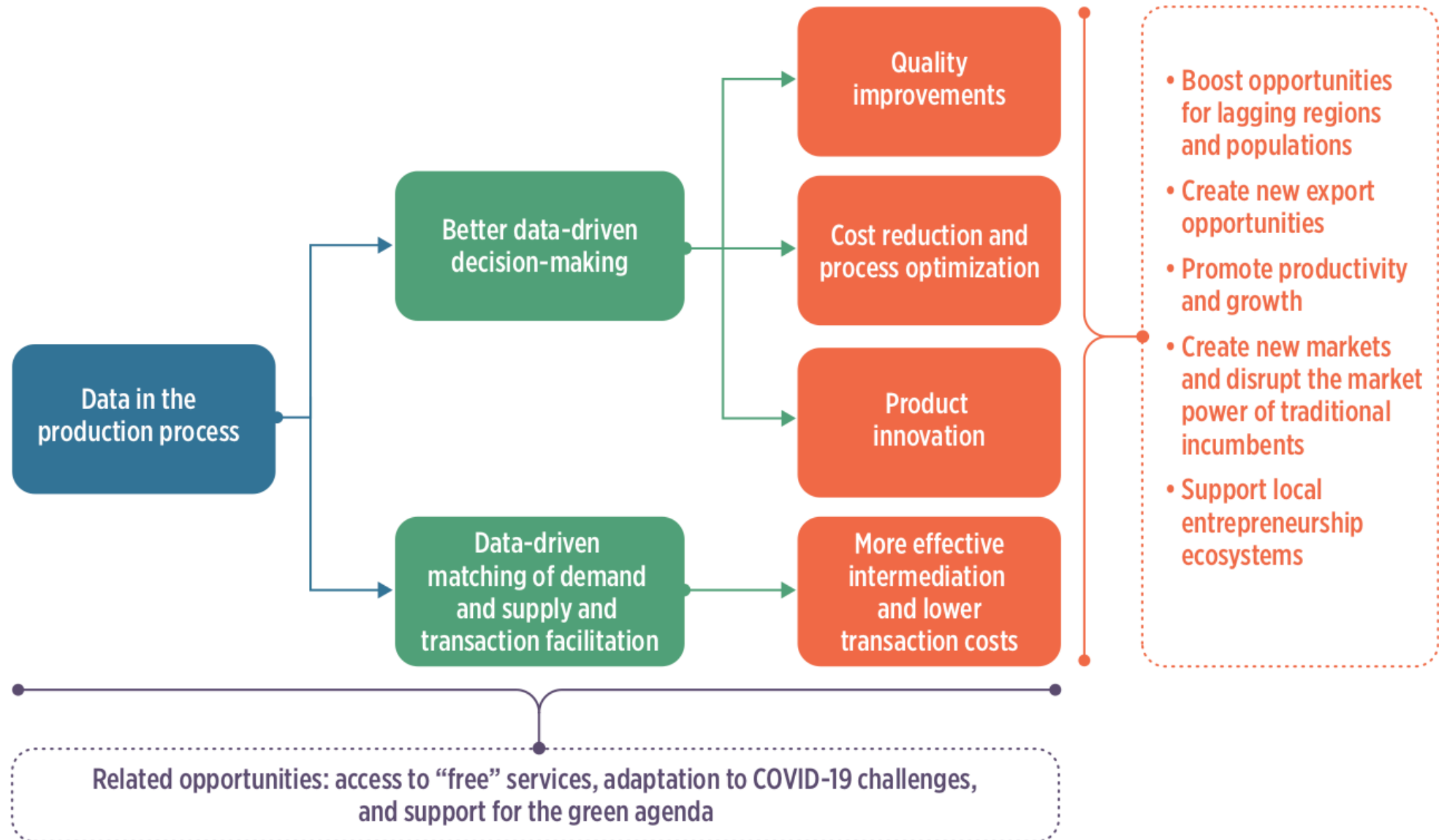
- Data not immune to influence from stakeholders
- Data not stored securely - too common in government and private sector databases
- Data not properly de-identified

Why gaps persist in public intent data

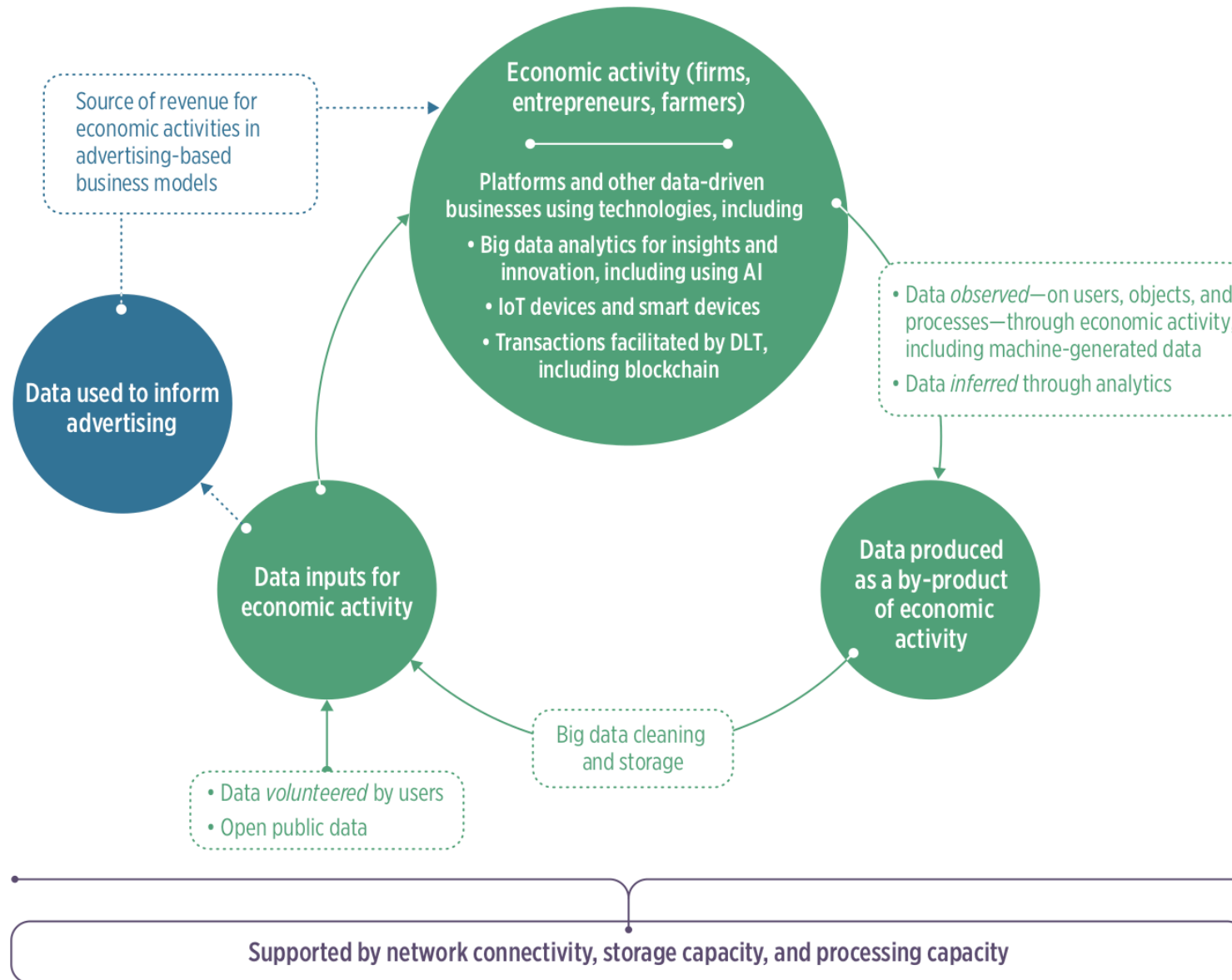
- Deficiencies in financing
- Deficiencies in technical capacity
- Deficiencies in governance
- Deficiencies in data demand

Private intent data

Data in the production process



Role of data in economic activity



Data inputs for economic activity

- Data collection by firms - “digital footprint”
- Use of public intent data by businesses

Sectors transformed by use of data in the production process

- Finance
- Agriculture
- Health
- Education
- Transport and logistics

Finance sector

- **Alternative credit scoring algorithms** - use of an individual's digital footprint to train machine learning algorithms to identify, score, and underwrite credit.
- **Payment systems** - digital payments; mobile payments.
- **Use of transaction data for product development** - Mastercard's Tourism Insights service allows leveraging big data to provide information on travellers' preferences.
- **Distributed ledger technology** - blockchain; eliminates financial intermediaries; embedding rules into smart contracts simplifies negotiation and verification processes.

Agriculture sector

- Remote sensing and geographic information systems - analysis of these data provide insights into farming operations; smart farming
- Internet-of-things (IoT) and embedded devices for monitoring farm and livestock conditions and farm operations
- Use of data to provide a range of services along the agriculture value chain
- Use of data to improve food traceability

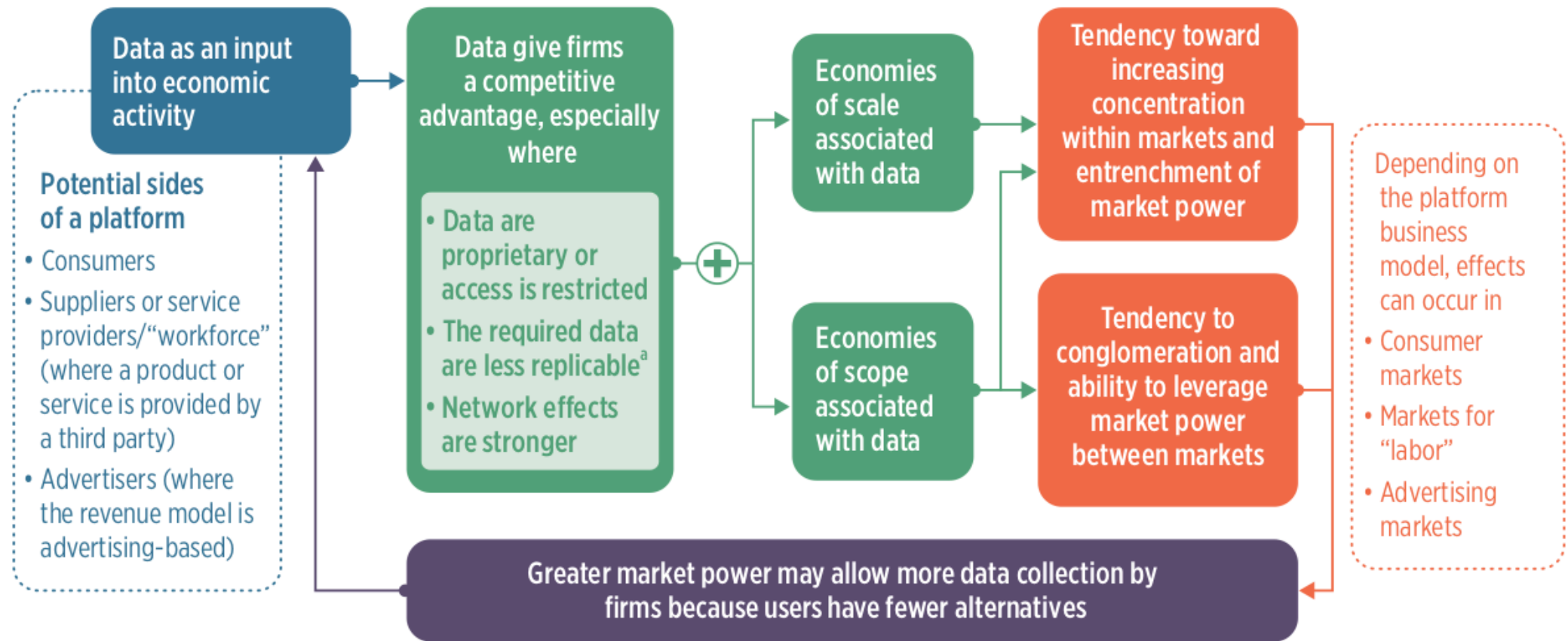
Education

- video conferencing/video communication tools
- digital supplemental learning platforms
- intelligent adaptive education

Transport and logistics sector

- Digital freight matching
- Digital courier logistics
- IoT-enabled cold chain

Risks to market structure from platform firms



Risks and adverse outcomes of data-driven businesses

- Increase propensity of dominant firms emerging
- Potential for exploitation of individuals
- Potential to increase inequality within and among countries